A WORKBENCH FOR FREECAD

InCad

A familiar Autodesk Inventor and Alibre style interface on top of the FreeCAD kernel

Test build 0.3 · macOS on Apple Silicon · FreeCAD 1.1

WHY

FreeCAD's kernel is strong. Its interface feels foreign if you come from Inventor

A new interface

Ribbon, model browser, dialogs and menus as in Inventor and Alibre

The same kernel

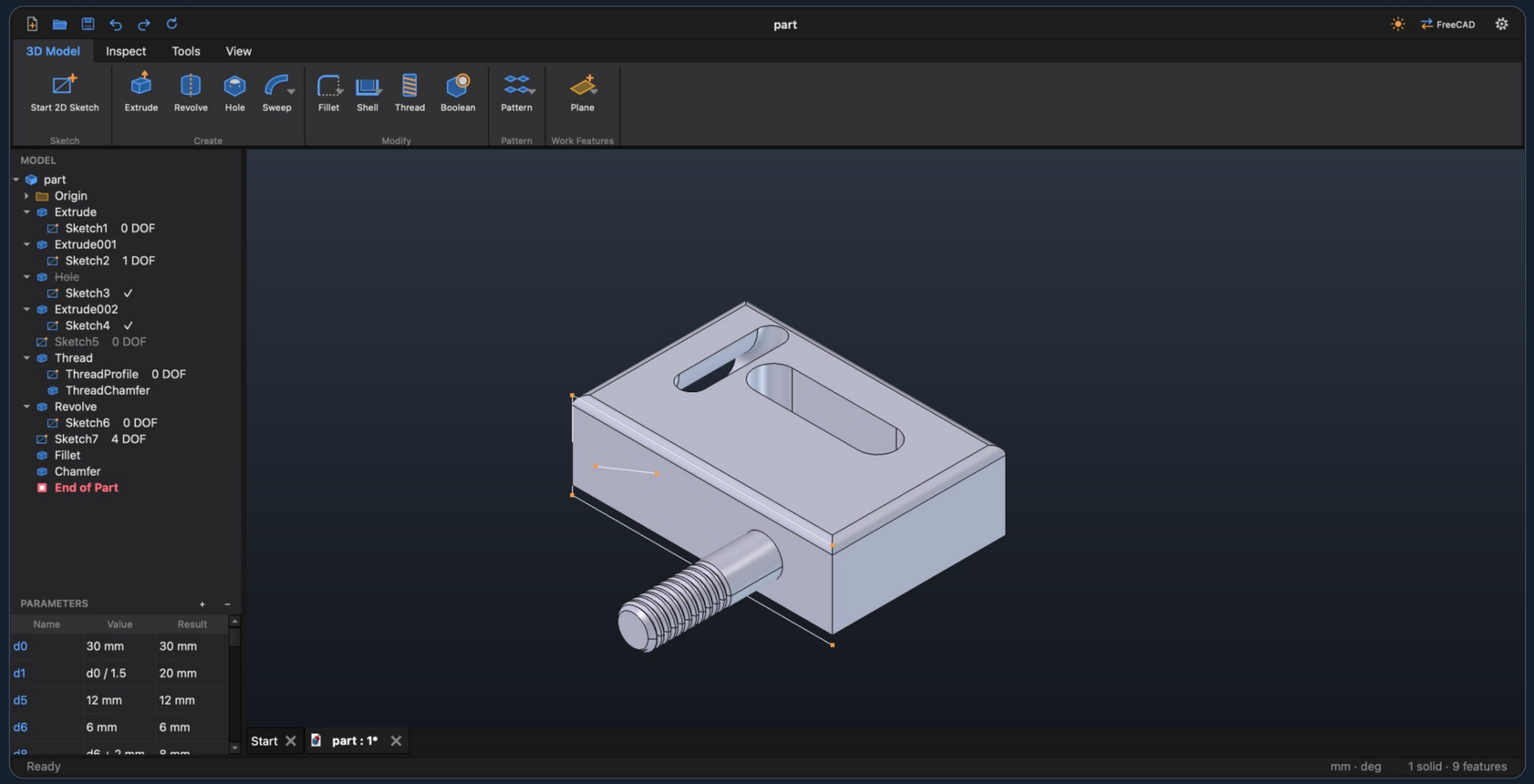
Under the hood: FreeCAD's own Sketcher and PartDesign

Ordinary files

Parts are saved as .FCStd and open in any FreeCAD

A FAMILIAR LAYOUT

This is how it looks



The ribbon on top, the model browser and parameters on the left, the model in the middle. Navigation on the right.

InCad with a real part: a block with a threaded shaft, slots and rounds.

WHAT'S INSIDE

Four pillars of the interface

Command ribbon

3D Model, Inspect, Tools and View tabs, and tabs of its own in a sketch

Model browser

One history per part, sketches under their features, End of Part

Dialogs with preview

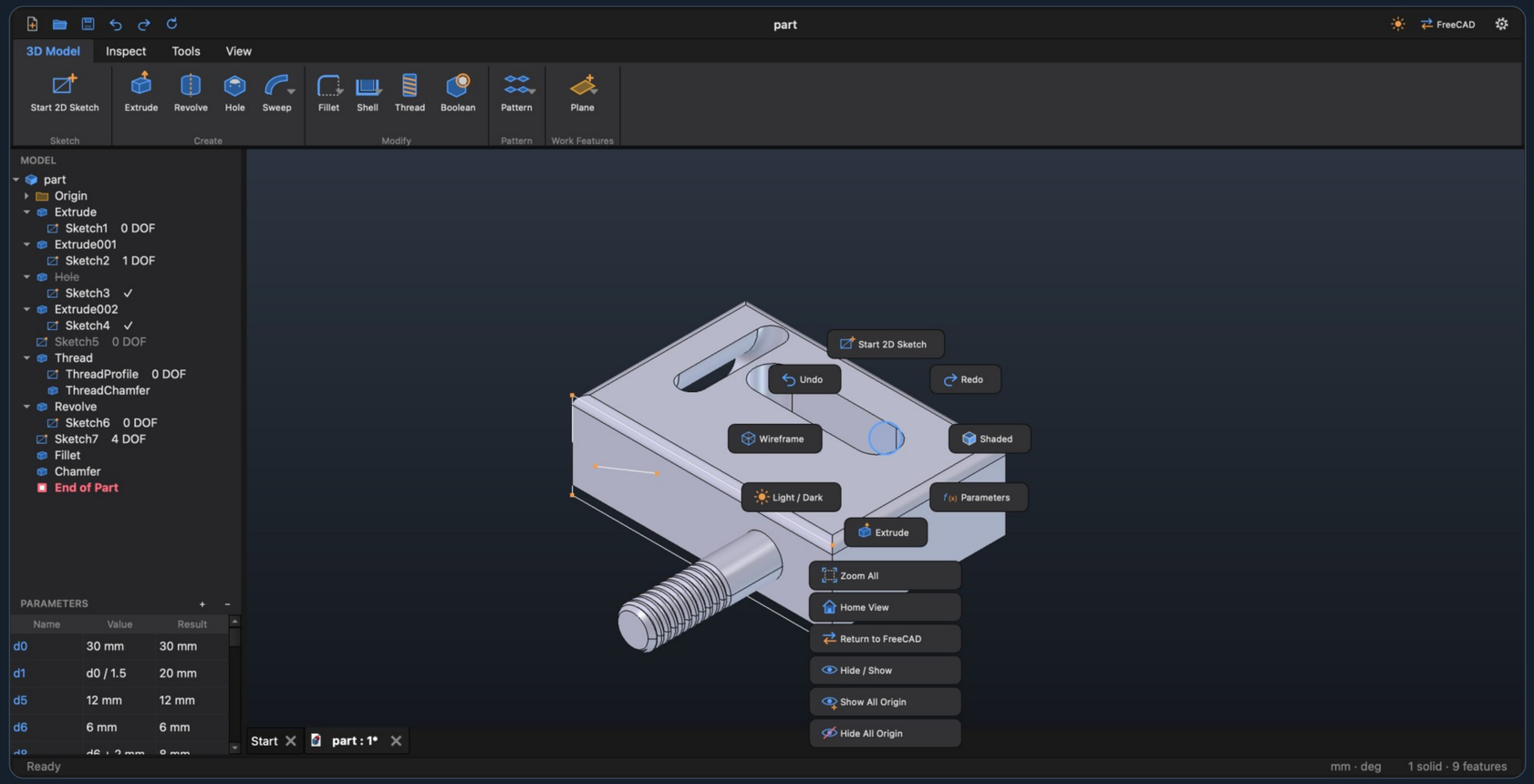
See the result at once, ▲▼ step the numbers, Undo inside the dialog

Marking menus

The right button opens a radial menu, a flick of the mouse picks the command

RIGHT MOUSE BUTTON

Marking menus

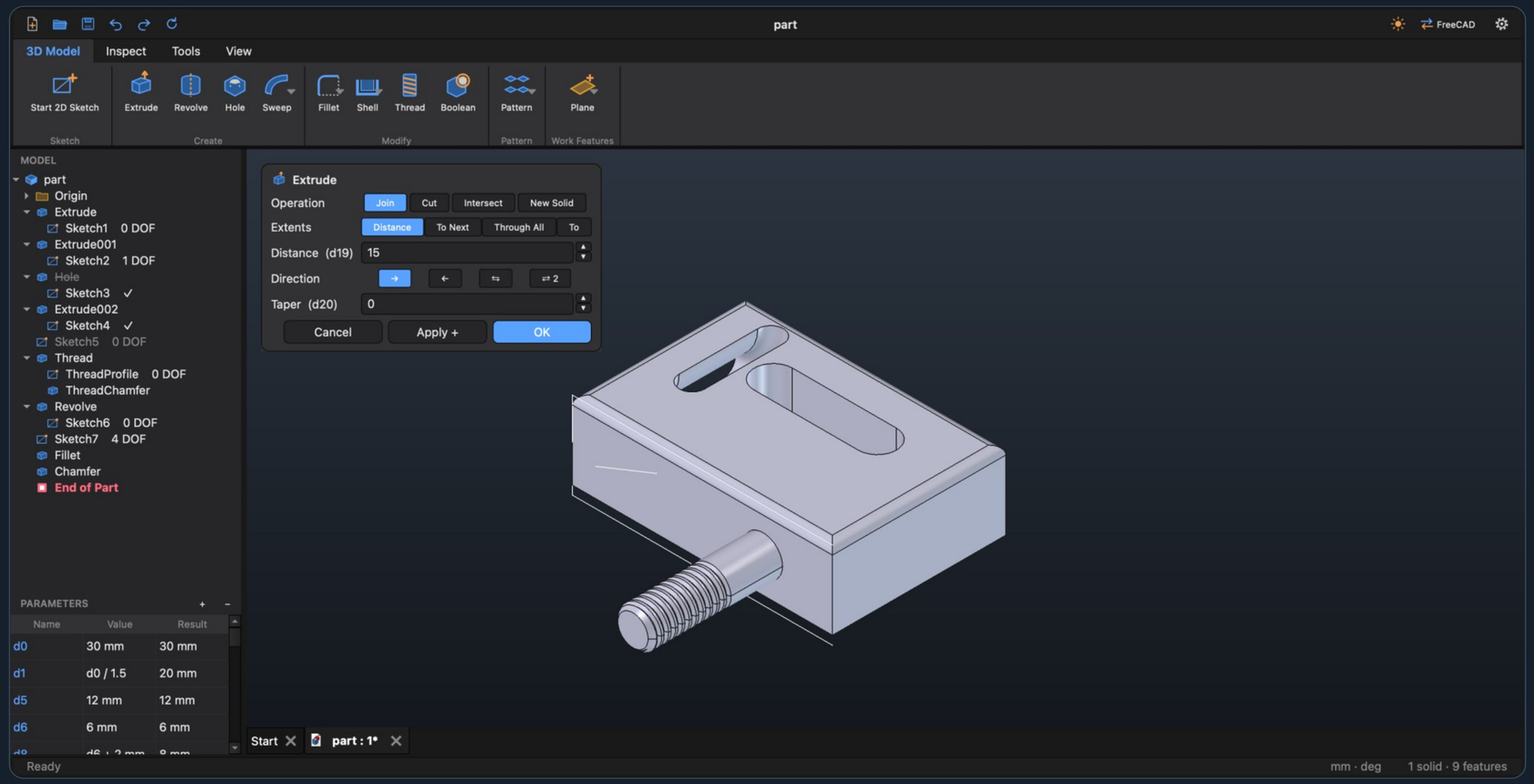


Commands around the ring are picked with a flick of the mouse, frequent actions are listed below.

Near a screen edge the menu moves inward; the gesture still starts where you pressed.

EDITING AN EXTRUDE

Feature dialog



Every change shows on the model at once. The ▲▼ arrows step the number, Shift steps by 10.

The (d19) label means the value is a model parameter you can use in formulas.

MODEL BROWSER

One history, as in Inventor

- **Every solid in one list** in the order made, rows of other solids name theirs
- **One End of Part**: drag the marker and the whole part rolls back to that point
- **Highlight both ways**: a feature in the browser shows its faces, a face in the view shows its feature

IN THE MENU

Show in Browser

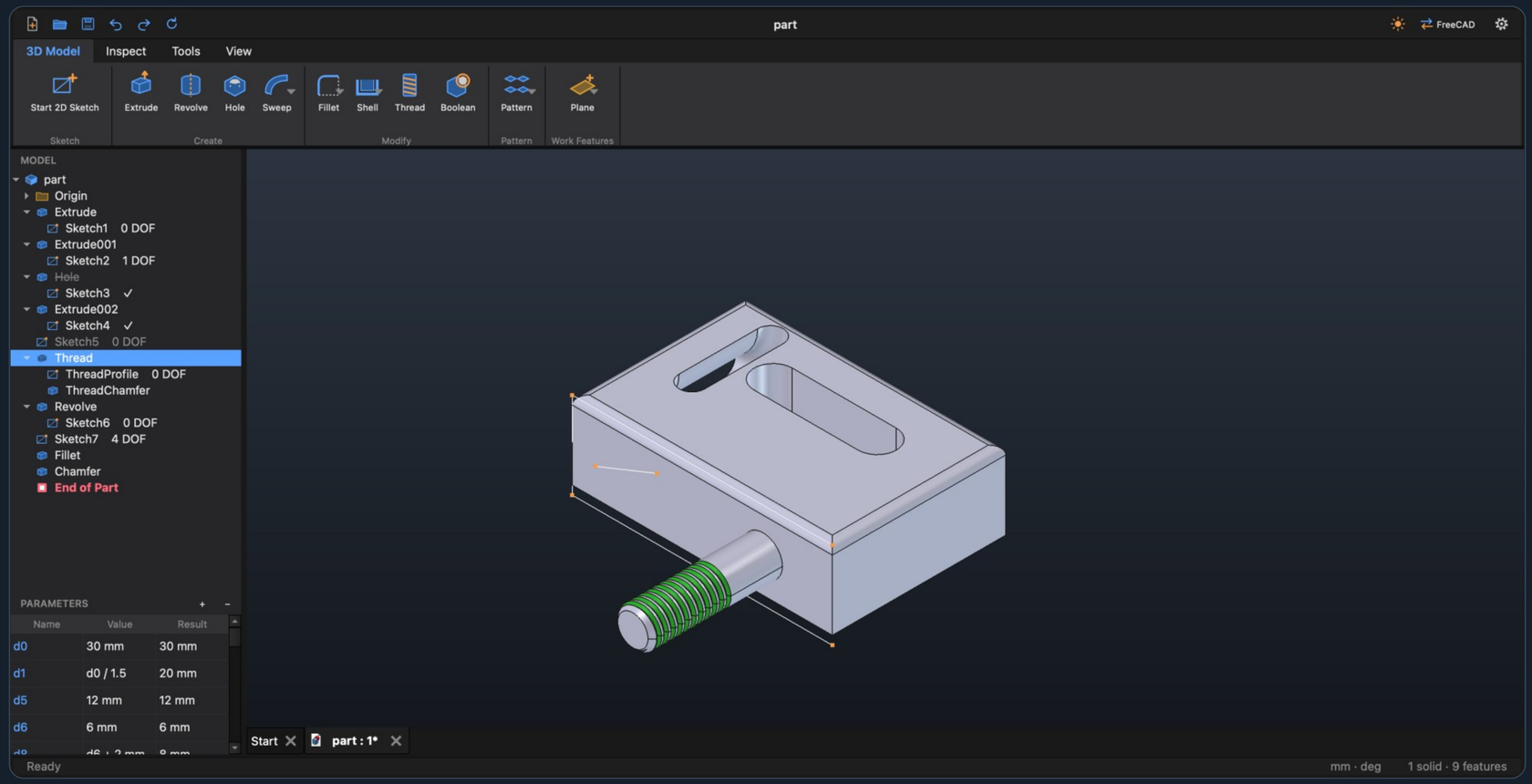
from the view: open the browser at the feature of the picked face

Show in Window

from the browser: light the feature's faces and zoom to them

BROWSER ↔ VIEW

Feature highlight



The Thread row in the browser lights the faces the thread made on the model.

It works the other way too: a click on a face lights its feature in the browser.

SKETCHES AND PARAMETERS

Named dimensions and formulas

Parameters

Each dimension gets a name, d0, d1… Write d0 / 1.5 and change everything from the Parameters panel

Patterns in a sketch

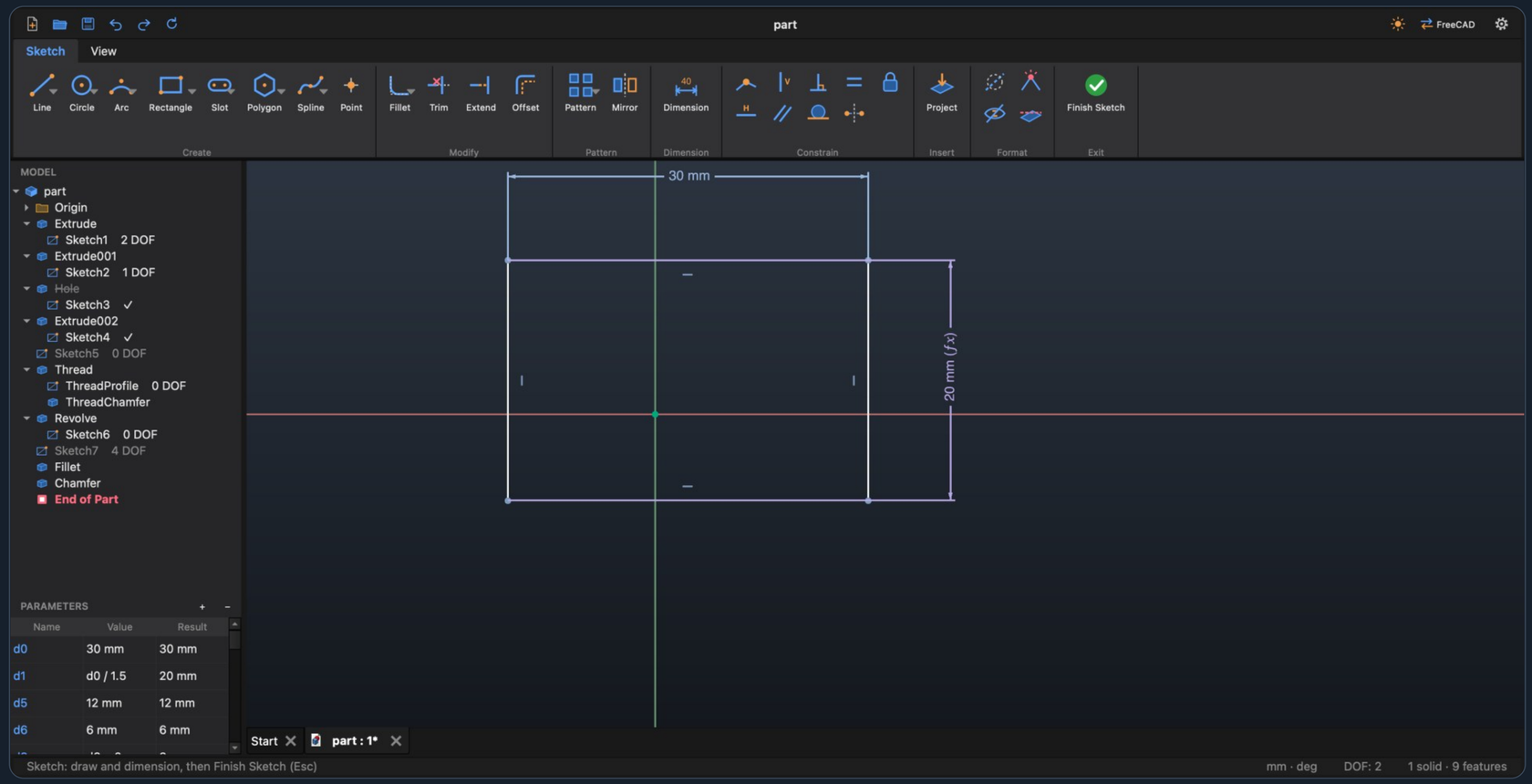
Rectangular, circular and along a path, with equal spacing between limits

Open contours

Paths for Sweep and Loft stay visible with their ends and points

A RIBBON OF ITS OWN FOR SKETCHING

Sketch mode



Drawing, editing, patterns, dimensions and constraints sit on one Sketch tab.

The 20 mm dimension is a formula (fx): d0 / 1.5 from the Parameters panel.

FEATURES

Everything for a part in one ribbon

Group	Commands
Create	Extrude, Revolve, Hole (with Chamfer), Sweep, Loft, Coil
Modify	Fillet, Chamfer, Shell, Thread, Draft, Boolean (Keep tool), Split, Threads Off / On
Patterns	Rectangular and Circular Pattern, Mirror
Work features	Plane, Axis, Point
Assemble	Make Assembly

Sweep with a wall thickness: pipes and ducts in one feature, one end or both open

NO SURPRISES

InCad takes care of what trips you up in FreeCAD

Separate solids

A box drawn away from the part becomes a solid of its own

The right solid

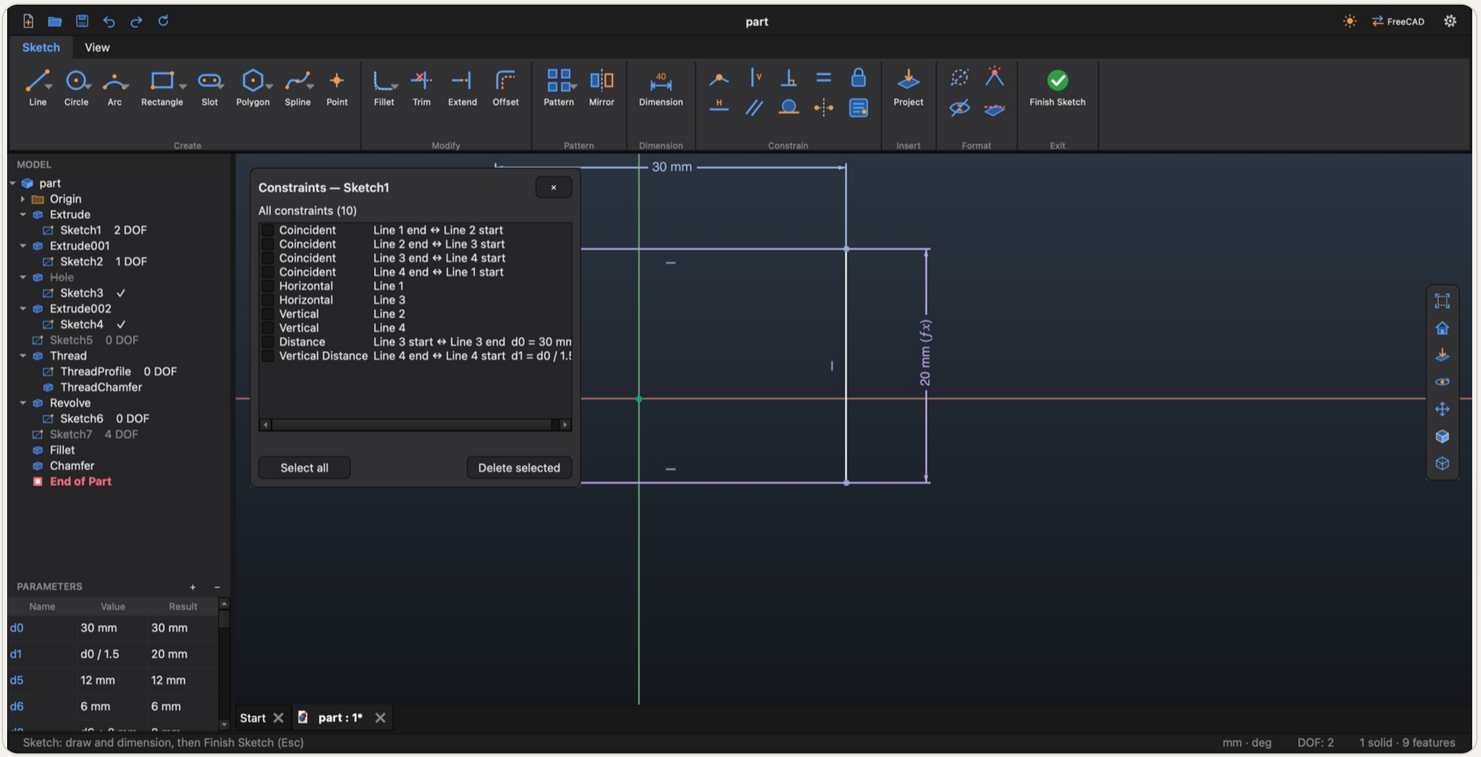
A sketch on a solid's face, and a Cut made from it, go to that solid

No freezes

A risky Loft is tried on a copy first instead of freezing FreeCAD

SKETCH

Constraint List



Every constraint of the sketch in one list, also the ones with no icon to click.

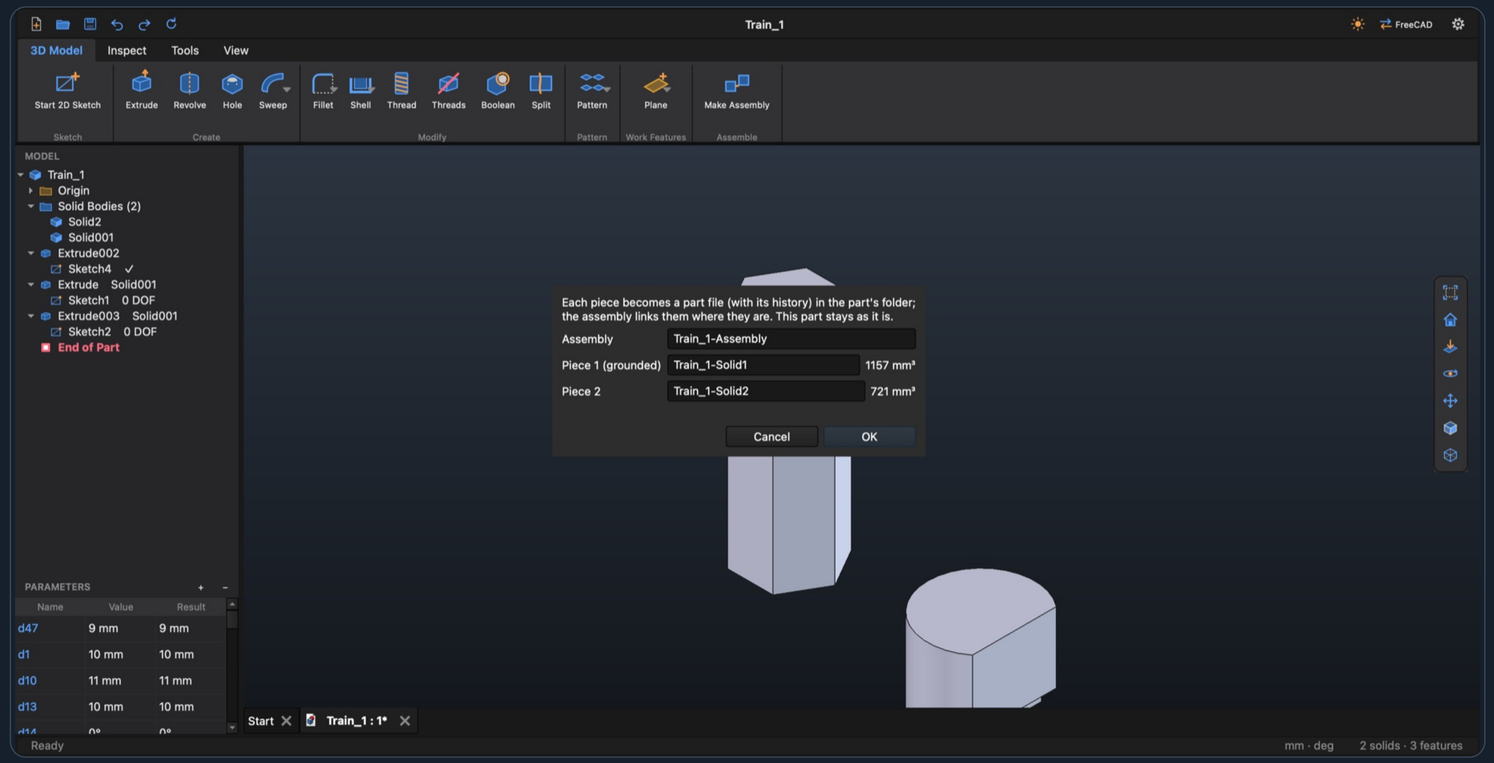
Pick a line or point to see only its own. Check rows and press Delete selected.

Or right-click a line or point and delete one of its constraints.

Projected edges are normal lines by default; Construction switches them.

NEW IN PARTS

Split and Make Assembly



SPLIT

The lines of a sketch cut the part. Keep all: each piece is a Solid. Keep picked: only the pieces you click stay.

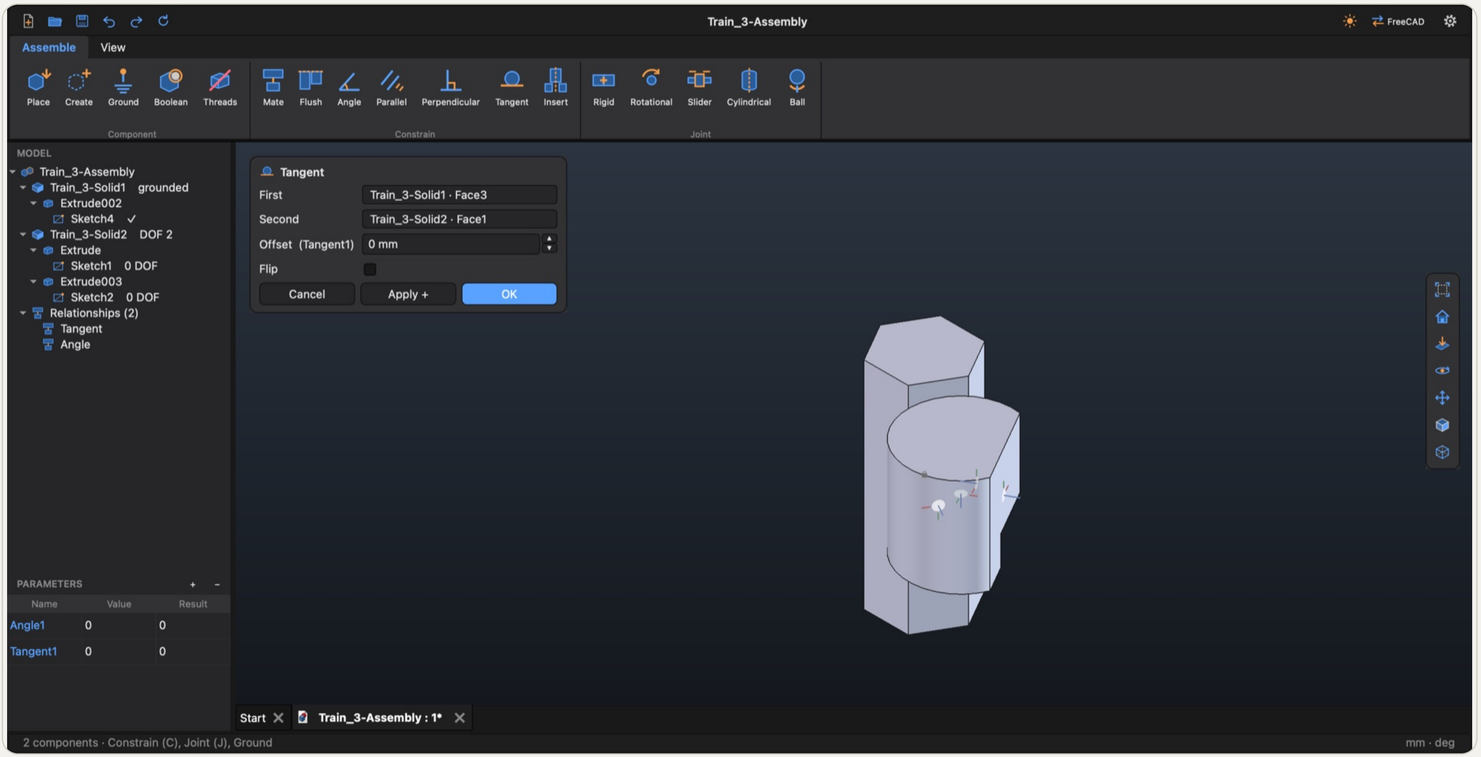
MAKE ASSEMBLY

Each separate piece becomes a part file with its history; a new assembly links them where they were.

Run it again and choose Replace existing files: the old files are overwritten.

ASSEMBLIES

Constraints and joints



CONSTRAINTS

Mate, Flush, Angle, Parallel, Perpendicular, Tangent, Insert

JOINTS

Rigid, Rotational, Slider, Cylindrical, Ball

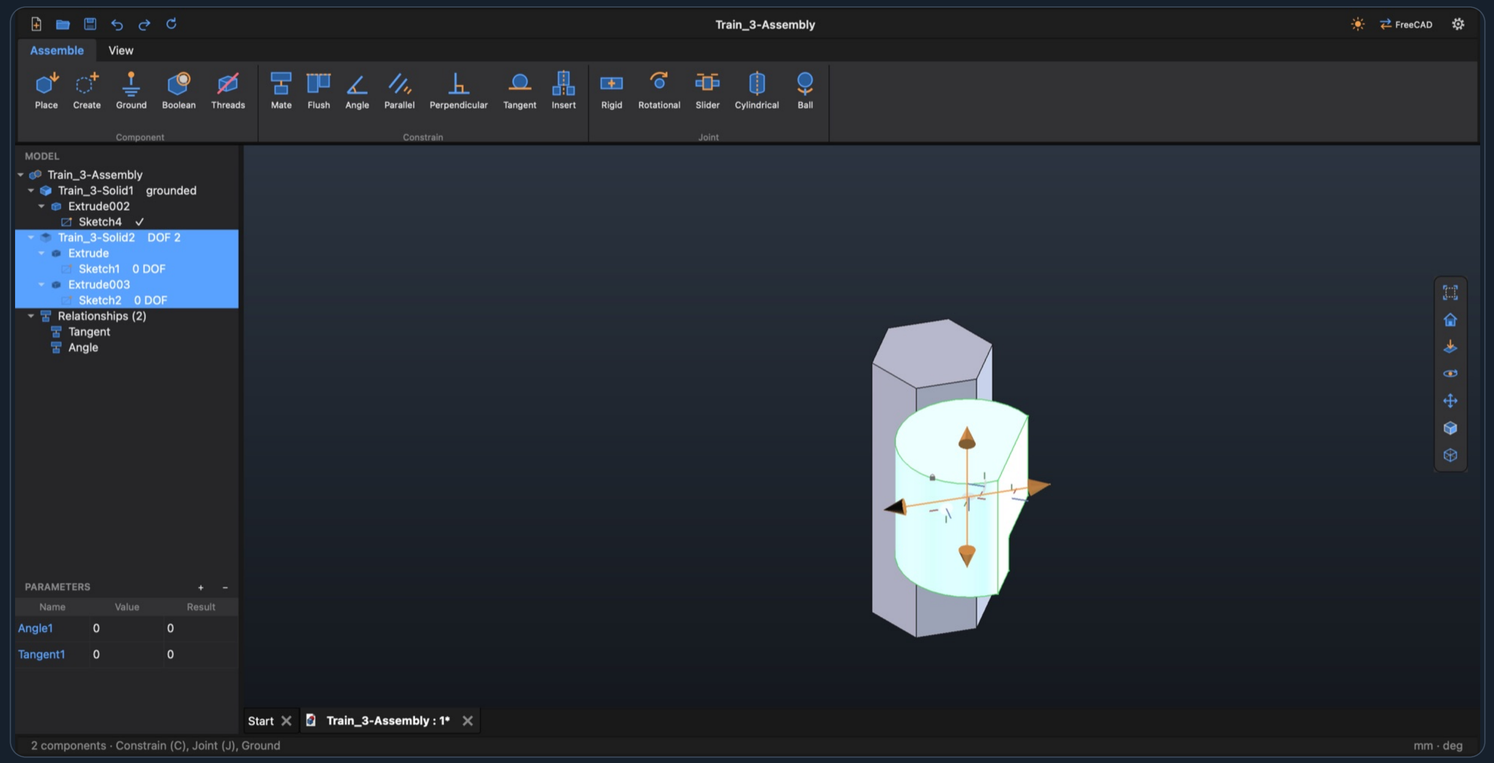
Flip turns the part over in the preview.

RELATIONSHIPS

Double-click a row to edit it. Right-click: Edit, Delete, Suppress, Rename.

ASSEMBLIES

Moving parts



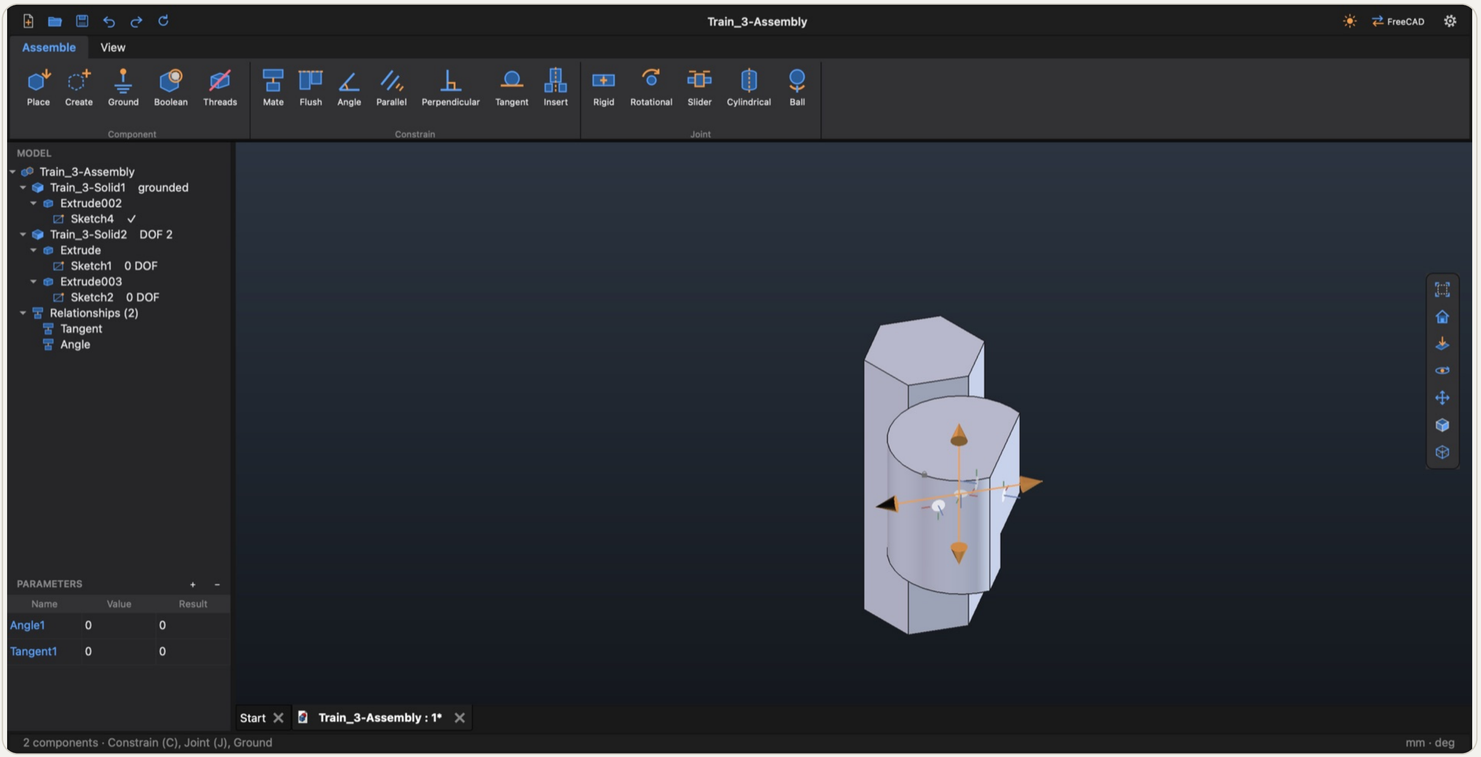
Drag a free part, or click it to get its gizmo: arrows slide, arcs turn.

Only along the part's own degrees of freedom. The other parts stay put.

One drag is one undo step.

ASSEMBLIES

Degrees of freedom



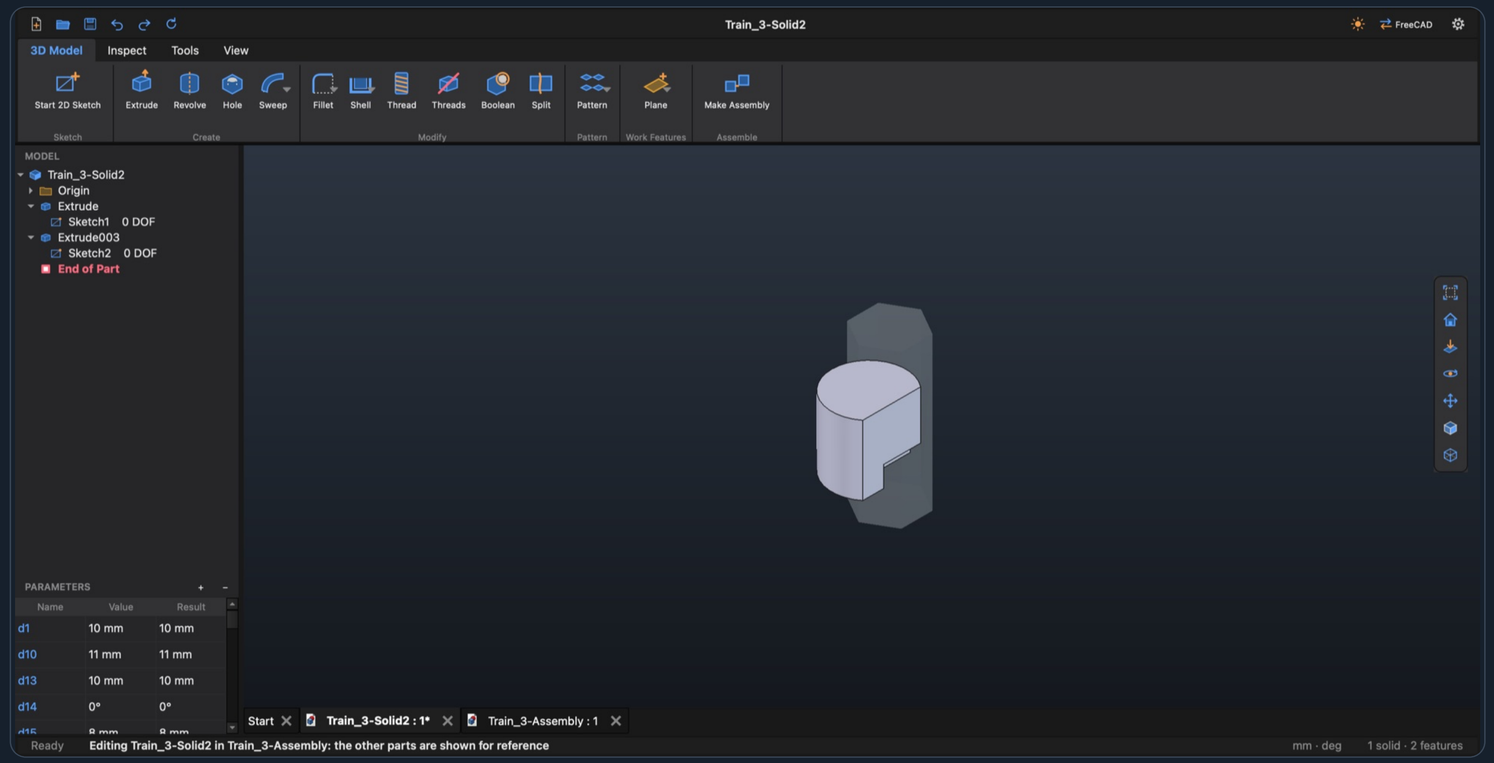
Each part's row shows Grounded or DOF n: the ways it can still move.

View › Degrees of Freedom draws Inventor's DOF symbols on the parts.

With it off, the part under the pointer shows its symbol as a tip.

ASSEMBLIES

Edit a part in context



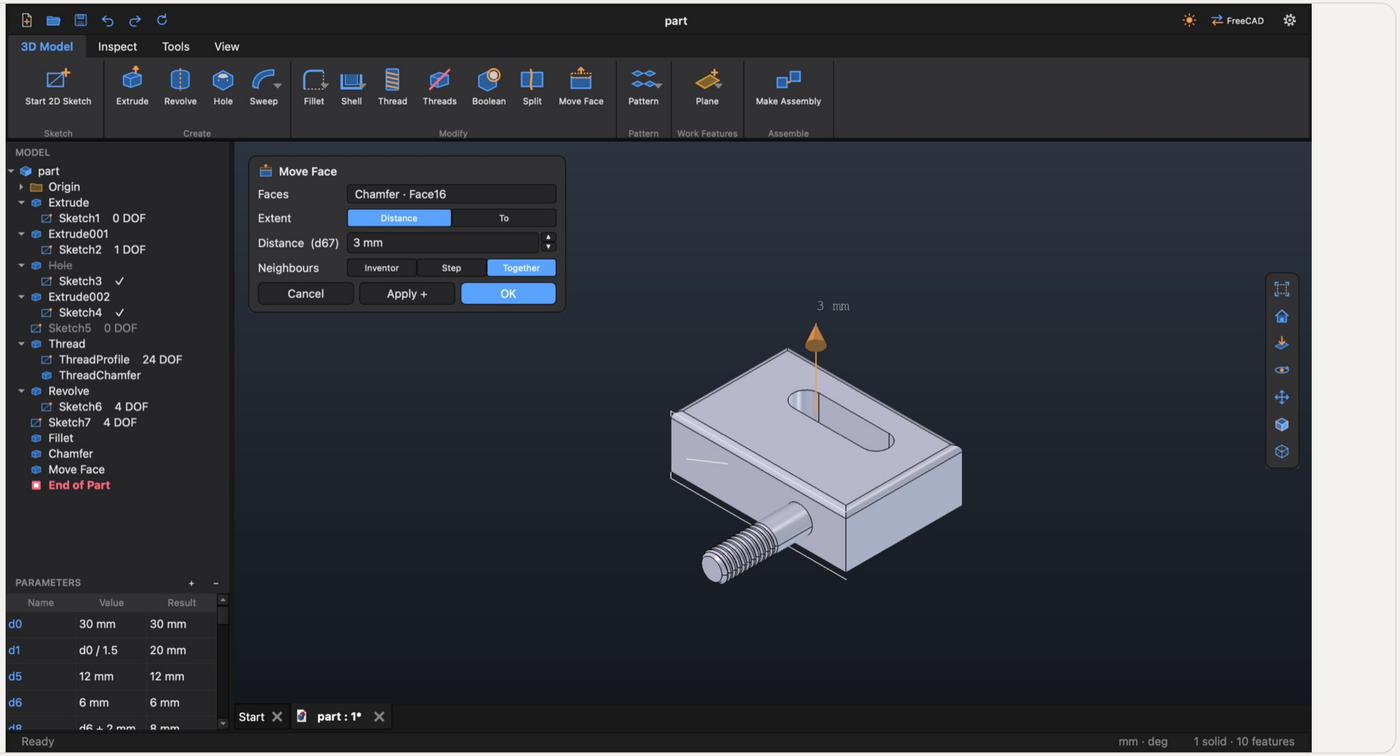
Edit opens the part in its tab with the other parts semi-transparent around it.

View › Assembly Context turns them off and on.

Back in the assembly, InCad asks before updating the part.

DIRECT EDITING

Move Face



Pick flat faces and push or pull them: type a distance, or drag the arrow and read the value at its tip.

Neighbours: Inventor (chamfers keep their angle), Step, or Together (the face moves with its chamfers and fillets).

To a parallel face, plane, edge or point. A step in the history: edit, suppress, delete. Finish move on top of the right-click menu.

BUILD STATUS

Early test version 0.3

What	Status
Parts	Tested, waiting for your feedback
Assemblies	Ready to test (new in 0.2)
Move Face (direct editing)	Ready to test (new in 0.3)
macOS, Apple Silicon	Supported
FreeCAD	1.1, tested with 1.1.3
Windows, Linux, Intel Macs	Later, if there is demand

Save often and try it on copies of important files

INSTALL

Four steps with the Addon Manager

1. Unzip the archive and copy the path of the InCad folder: Option+Cmd+C
2. Settings › Addon Manager › Custom repositories › “+”, paste the path
3. Tools › Addon Manager › InCad › Install
4. Restart FreeCAD and choose the InCad workbench

To start FreeCAD in InCad: Settings › Workbenches › Available Workbenches › Startup › Default workbench: InCad

Details and the manual install: README.txt in the archive. The full manual: InCad-manual.pdf

HOW TO HELP

Found a bug? Send four things

- **Steps:** what you did and what you expected
- **Log:** the file ~/Library/Logs/InCad.log
- **Model:** the .FCStd file, if you can share it
- **Versions:** build name, macOS and FreeCAD; after a crash, macOS's report too

Try it and tell us what's wrong

Every report brings InCad closer to the CAD you are used to

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